PRACTICAL EXERCISE 26

Providing structured data to an AI tool for automated statistical analysis and metric extraction.

USER PROMPT

"Analyze this student marks dataset. Calculate the average marks, identify the highest and lowest marks, count students scoring below 40, and summarize the results in a table. Explain any assumptions."

1. THE PROVIDED DATASET

A simulated CSV dataset of student marks was provided to the AI along with the prompt.

Student ID	Name	Marks
101	Alice Walker	85
102	Bob Chen	38
103	Charlie Smith	92
104	Diana Kim	45
105	Ethan Jones	35

2. AI GENERATED ANALYSIS

59.0	92	35	2
AVERAGE MARKS	HIGHEST SCORE	LOWEST SCORE	STUDENTS < 40

AI Stated Assumptions:

- The dataset is complete, and there are no missing or null values in the "Marks" column.
- The marks are represented as raw numerical scores (out of a maximum of 100).
- A score strictly below 40 is considered failing for the purpose of the count.

3. FOLLOW-UP: RESULT VERIFICATION

Verification Task: It is critical to independently verify mathematical calculations performed by Large Language Models, as they can occasionally hallucinate arithmetic. Below is a Python script using the pandas library to verify the AI's claims.

```
import pandas as pd # Reconstruct the dataset data = { 'Name': ['Alice', 'Bob', 'Charlie', 'Diana', 'Ethan'], 'Marks': [85, 38, 92, 45, 35] } df = pd.DataFrame(data) # Execute mathematical operations print(f"Average Marks: {df['Marks'].mean()}") print(f"Highest Mark: {df['Marks'].max()}") print(f"Lowest Mark: {df['Marks'].min()}") print(f"Count < 40: {len(df[df['Marks'] < 40])}")
```

Python Console Output

```
Average Marks: 59.0
Highest Mark: 92
Lowest Mark: 35
Count < 40: 2
```

Key Learning Outcomes

1. Automated Summarization: AI tools can rapidly ingest small CSV files or pasted tables and extract standard descriptive statistics accurately, formatting them into readable summaries.

2. State Assumptions: Prompting the AI to "explain assumptions" is a critical habit. It reveals how the model interpreted the data (e.g., assuming no null values) which impacts the validity of the results.

3. Trust but Verify: While modern AI is generally accurate with small datasets, complex aggregation requires independent verification via code (Python/R) or traditional spreadsheet software (Excel).